# Least squares regression line

**1** Any decimal between $0.8$ and $0.99$.

**2**

**a** $y = 19 - 2x$

**b** $y = -5 + 2x$

**3** Weak and positive association.

**4**

**a**

**i** $-0.85$

**ii** Strong negative linear relationship

**iii** $y = -0.1x + 30.3$

**b**

**i** $0.75$

**ii** Moderate positive linear relationship

**iii** $y = 5.8x - 99.0$

**5**

**a** $n = -2.50t + 20.94$

**b** $r = -0.89$

**c** As the temperature increases, the number of hot cocoa sold will decrease.

**6**

**a** $r = 0.86$

**b** Strong positive linear relationship

**c** $y = 0.89x + 2.78$

**d** $25.03^\circ\text{C}$

**e** $32^\circ\text{C}$, because the other two values are examples of extrapolation.

**7**

**a** $r = -0.908$

**b** Strong negative linear relationship

**c** $y = -0.06x + 6.37$

**d** $6.07\ \text{kg}$

**e** Reliable due to interpolation and a strong correlation.

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**8**

a $r = 0.86$

b Strong positive linear relationship

c $y = 0.98x - 7.94$

d $\$50.86$

e Reliable due to interpolation and a strong correlation.

**9**

a $r = -0.91$

b Strong negative linear relationship

c $y = -6.6x + 80.1$

d $47.1$

e Reliable due to interpolation and a strong correlation.

**10**

a $r = 0.67$

b Moderate positive linear relationship

c $y = 0.21x - 2.09$

d $15$ computers

e Very unreliable due to extrapolation and a moderate to weak correlation.

**11**

a $r = 0.79$

b Strong positive linear relationship

c $y = 0.025x - 550.108$

d $\$1449.89$

e $\$80\,000$ because the other two are examples of extrapolation.

**12**

a $r = 0.997$

b Strong positive linear relationship

c $y = 0.85x + 1.28$

d $52.28\%$

e Reliable due to interpolation and a strong correlation.

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**13**

**a** $r = -0.20$

**b** $(1, 1.28)$

**c** $r = 0.79$

**d** Strong positive linear relationship

**e** $y = 0.05x + 0.23$

**f** $1.23$

**g** Despite a moderate correlation, unreliable due to extrapolation.

## Calculating residuals

**14**

**a**

| $x$ | $y$ | $\hat{y}$ | Residual |
|---|---|---|---|
| 10 | 29 | 25 | 4 |
| 14 | 29 | 37 | $-8$ |
| 7 | 20 | 16 | 4 |
| 6 | 21 | 13 | 8 |
| 20 | 58 | 55 | 3 |
| 13 | 26 | 34 | $-8$ |

**b**

| $x$ | $y$ | $\hat{y}$ | Residual |
|---|---|---|---|
| 12.8 | $-32.7$ | $-33.3$ | 0.6 |
| 11.4 | $-30.3$ | $-30.3$ | 0 |
| 11.1 | $-29.5$ | $-29.7$ | 0.2 |
| 6.5 | $-21.5$ | $-20.1$ | $-1.4$ |
| 10 | $-28.5$ | $-27.4$ | $-1.1$ |

**15**

**a**

| $x$ | 1 | 3 | 5 | 7 | 9 |
|---|---|---|---|---|---|
| $y$ | 22.7 | 22.3 | 24.2 | 21.8 | 21.5 |
| $\hat{y}$ | 25.2 | 23.4 | 21.6 | 19.8 | 18 |
| Residuals | $-2.5$ | $-1.1$ | 2.6 | 2 | 3.5 |

**b**

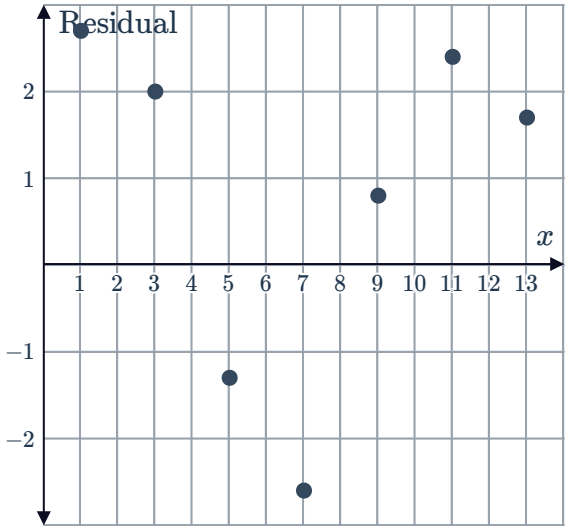
| $x$ | 5 | 6 | 7 | 8 | 9 |
|---|---|---|---|---|---|
| $y$ | 37.7 | 37.2 | 21.1 | 27.1 | 44 |
| $\hat{y}$ | 28.9 | 30.4 | 31.9 | 33.4 | 34.9 |
| Residuals | 8.8 | 6.8 | $-10.8$ | $-6.3$ | 9.1 |

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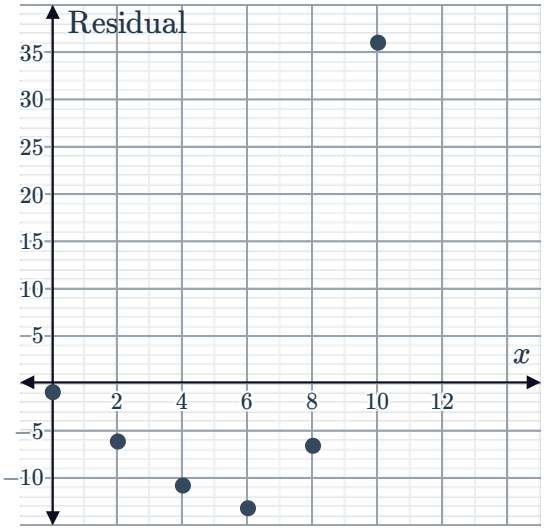
# Residual plots

**16**

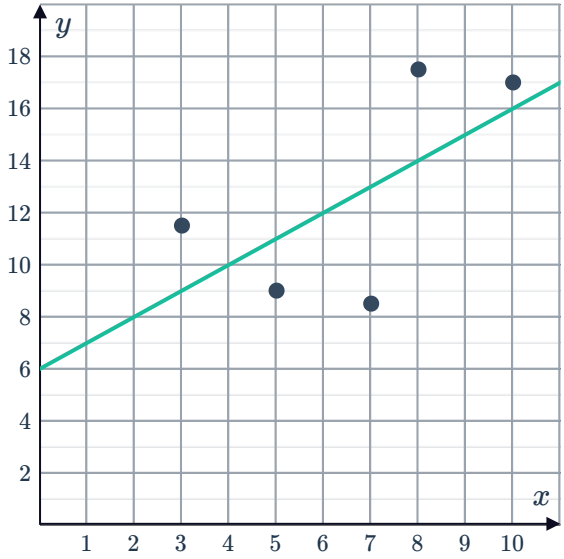
**a**




**b**




**17**




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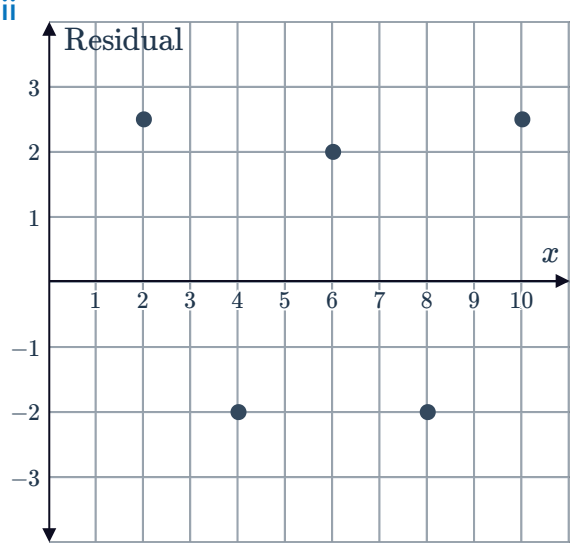
**18**

**a**

**i**

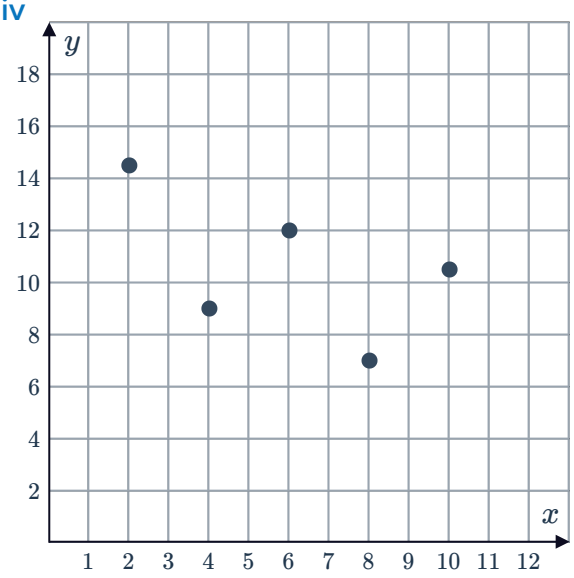
| $x$ | $y$ | $\hat{y}$ | Residuals |
|---|---|---|---|
| 2 | 14.5 | 12 | 2.5 |
| 4 | 9 | 11 | $-2$ |
| 6 | 12 | 10 | 2 |
| 8 | 7 | 9 | $-2$ |
| 10 | 10.5 | 8 | 2.5 |

**ii**




**iii** Linear relationship

**iv**



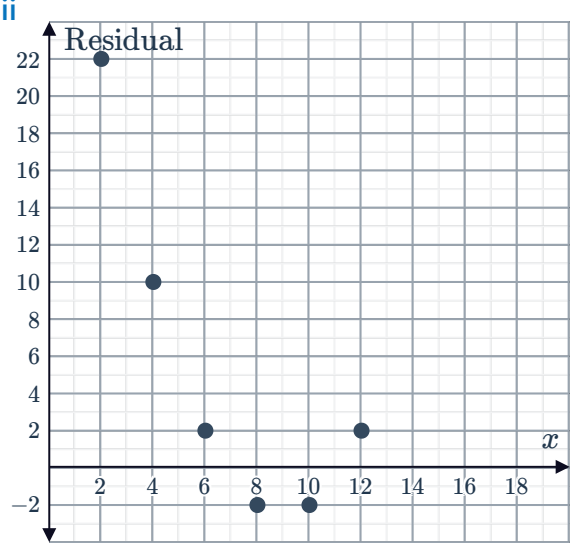

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**b**

**i**

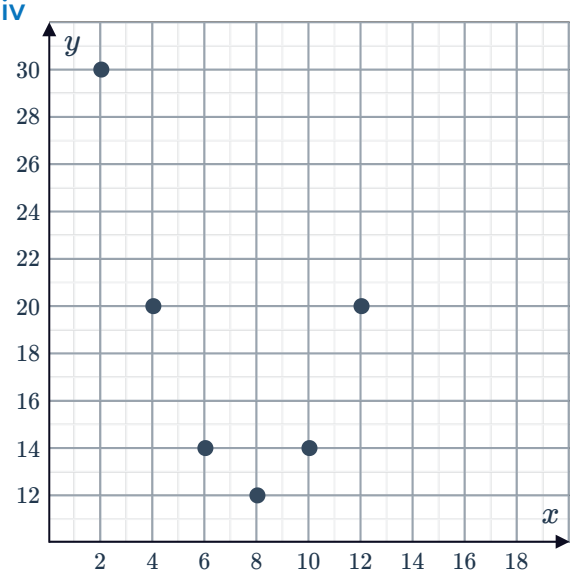
| $x$ | $y$ | $\hat{y}$ | Residuals |
|---|---|---|---|
| 2 | 30 | 8 | 22 |
| 4 | 20 | 10 | 10 |
| 6 | 14 | 12 | 2 |
| 8 | 12 | 14 | −2 |
| 10 | 14 | 16 | −2 |
| 12 | 20 | 18 | 2 |

**ii**




**iii** Non-Linear relationship

**iv**



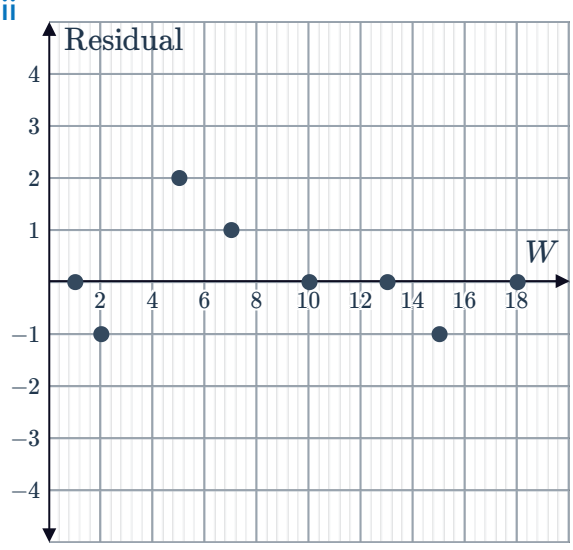

**19**

**a**

**i**

| $W$ | $C$ | Model value | Residual |
|---|---|---|---|
| 1 | 7 | 7 | 0 |
| 2 | 9 | 10 | −1 |
| 5 | 21 | 19 | 2 |
| 7 | 26 | 25 | 1 |
| 10 | 34 | 34 | 0 |
| 13 | 43 | 43 | 0 |
| 15 | 48 | 49 | −1 |
| 18 | 58 | 58 | 0 |

**ii**




**iii** Yes, the model is a good fit.

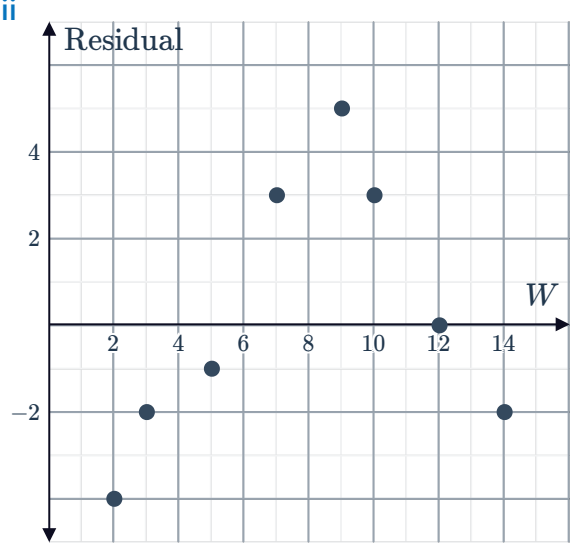
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**b**

**i**

| $W$ | $R$ | Model value | Residual |
|---|---|---|---|
| 2 | 7 | 11 | $-4$ |
| 3 | 12 | 14 | $-2$ |
| 5 | 19 | 20 | $-1$ |
| 7 | 29 | 26 | 3 |
| 9 | 37 | 32 | 5 |
| 10 | 38 | 35 | 3 |
| 12 | 41 | 41 | 0 |
| 14 | 45 | 47 | $-2$ |

**ii**




**iii** No, it is not a good fit.

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